

Top-Down Pooling and Anticipatory Exposure

Abstract

The standard signaling story explains separation from below: low types cannot afford to imitate high types. This paper identifies the mirror failure mode in a two-type dynamic psychological signaling model. Sending an ambitious signal increases current persuasive value, but it also creates a one-period reputational exposure: today's action can be repriced tomorrow if the judgment state turns to high scrutiny.

Each type has a primitive self-model accuracy level, and a higher true type can carry higher exposure risk under adverse repricing. The high-scrutiny transition probability is $q = \Pr(j_{t+1} = H \mid j_t = L)$. The current rent from an ambitious signal for type θ is

$$D_\theta^L(q) = B_\theta - \beta q \mu_H \Phi_\theta.$$

The key is that $\Phi_H > \Phi_L$ and $B_H > 0$ can coexist.

This creates top-down collapse. Let

$$q_{IC} = \frac{B_H}{\beta \mu_H \Phi_H}, \quad q_{D1} = \text{the D1 willingness threshold}, \quad q^* = \max\{q_{IC}, q_{D1}\}.$$

For $q > q^*$, a binary-signal pure strategy that separates is not self-confirming: the high type would withdraw from s_A even on-path, and D1 then removes the high-type interpretation of off-path s_A .

The paper first proves a fixed-point selection result in the binary class: the unique D1-consistent assessment is pooling on the safe signal, while the ambitious signal becomes a signaling dead-end from above. A microfounded dynamic section then shows this collapse is robust to trust-capital and reinforcement effects: they shift margins, but bounded endogenous drift cannot generate automatic restoration. A continuous-type extension in Section 7 describes the observable shadow as polarization of ambitious senders, not simple mean reversion.

1 Introduction

In standard costly signaling, separation is sustained from below: low types cannot afford high-cost signaling, so they do not mimic high types. The model here studies the opposite failure mode. A good type may still benefit from an ambitious signal today, but can rationally retreat because that action creates a one-period exposure that can be repriced against her future credibility.

The mechanism is dynamic and belief-driven. The ambitious action has two effects. It is informative in the current period, but it also claims something about the sender's self-model. If high scrutiny arrives tomorrow, the claim can be reinterpreted as evidence of weaker self-understanding and thus impose an additional penalty. This is adverse repricing, not a true-type audit.

The primitives are minimal: two types, two sender signals, and an exogenous binary judgment state. The sender observes the low-scrutiny risk $q = \Pr(j_{t+1} = H \mid j_t = L)$ before acting. If she sends the ambitious signal, tomorrow's state can impose an extra epistemic cost. If she sends the pooling-safe action, that one-period shadow disappears.

The model has three moving parts.

1. A static, one-period D1 core that shows a top-down reversal: high types exit separation first when q is large.
2. A fixed-point closure argument that combines sender ICs and D1 beliefs in a binary-signal pure-strategy psychological assessment.
3. A dynamic forward-invariance result that adds trust-capital/semantic dynamics and shows endogenous accumulation does not automatically repair the collapse.

The first stage identifies how adverse repricing plus a D1-style off-path map can select pooling from above. The second stage closes the belief loop in binary form: a self-referential interpretation where high occupancy of the ambitious signal implies high-type meaning, but then current IC may already fail, and off-path reinterpretation then collapses that meaning. The third stage asks whether time and dynamics can reverse this outcome.

The result is defensive: no new stopping phenomenon appears. Static and belief-loop collapse does not self-correct through learning or accumulation alone.

Section 2 introduces primitives and the one-period IC comparison. Section 3 derives the exposure ranking from primitive self-accuracy levels. Section 4 gives the static D1 core and the fixed-point threshold $q^* = \max\{q_{IC}, q_{D1}\}$. Section 5 closes the belief loop in binary strategies. Section 6 gives the dynamic forward-invariance result. Section 7 presents a continuous-type extension, where the surviving ambitious signal can become polarized.

2 Primal Structure and Benchmark

This paper studies a signaling environment with two sender types and two signals, where the same signal that can separate today also creates a near-term reputation risk that may be repriced tomorrow. The baseline is the dynamic extension of Spence logic needed for the fixed-point reversal.

2.1 Types, signals, and judgment states

There is a sender with type

$$\theta \in \{\theta_L, \theta_H\}, \quad \theta_H > \theta_L.$$

The receiver does not observe the type.

The sender chooses one of two signals

$$s \in \{s_A, s_P\}.$$

Signal s_A is the ambitious action, and s_P is the pooling-safe action.

Judgment intensity follows a public Markov state

$$j_t \in \{L, H\}.$$

State L is ordinary evaluation, and state H is high-scrutiny evaluation. Let

$$\mu_H > \mu_L > 0$$

denote the scrutiny multipliers and

$$q := \Pr(j_{t+1} = H \mid j_t = L)$$

be the transition probability from low to high. The sender observes q before choosing a signal.

2.2 One-period exposure shadow

Choosing s_A creates a one-period exposure state that can be repriced tomorrow. Let

$$X_t \in \{0, 1\}$$

indicate whether last period's action is exposed in the next period. If $s_t = s_A$, then $X_{t+1} = 1$; if $s_t = s_P$, then $X_{t+1} = 0$. In this baseline the exposure horizon is one period: choosing the safe signal today removes the next-period exposure created by a yesterday ambitious action.

If $X_{t+1} = 1$ and $j_{t+1} = H$, the sender incurs a future epistemic penalty proportional to a true-type exposure index. This is the anticipatory shadow that drives behavior in the low state.

2.3 Current payoff and the anticipatory rent

Let B_θ denote the current net separation surplus from s_A versus s_P in state L when the receiver interprets the ambitious signal in the high-type way. The low-state payoff gain from sending s_A is

$$D_\theta^L(q) = B_\theta - \beta q \mu_H \Phi_\theta,$$

where $\Phi_\theta > 0$ is the true-type exposure cost derived from primitive self-model accuracy (Section 3), and $\beta \in (0, 1)$ is the discount factor.

The type- θ IC threshold is

$$q_\theta^* := \frac{B_\theta}{\beta \mu_H \Phi_\theta},$$

whenever $B_\theta > 0$. Type θ is willing to send s_A in the low-scrutiny state iff

$$q < q_\theta^*.$$

2.4 Static benchmark

When $q = 0$, current risk from future repricing is absent. The model then collapses to a one-shot separating benchmark.

Assumption 1 (Static separation).

$$B_H > 0, \quad B_L < 0.$$

Under Assumption 1, with no anticipated repricing s_A is uniquely attractive only to the high type.

2.5 Anticipatory reversal object

The model's distinctive case is high-type exposure vulnerability. Even if $B_H > 0$, the same transition probability q can make high types abandon s_A because high types lose more from possible repricing in state H .

As the baseline, define

$$q_{IC} := \frac{B_H}{\beta \mu_H \Phi_H},$$

which is the IC threshold that would bind if s_A were still interpreted as a high-type signal.

Define also the D1-type willingness threshold

$$q_{D1} > 0$$

from the off-path comparison in Section 4.

The combined threshold is

$$q^* := \max\{q_{IC}, q_{D1}\}.$$

For $q > q^*$, the top-down reversal can be pinned down in static one-period terms: high types lose the static incentive to stay on s_A , and D1 no longer keeps s_A high-type in off-path beliefs.

3 Psychological Exposure and Bayesian Surprise

The benchmark environment in Section 2 treats the ambitious signal s_A as a costly signal of ability. This section adds the psychological object that distinguishes the model from standard costly signaling. The ambitious signal does not only reveal information about the sender's type. It also exposes the sender's self-model. A sender who chooses s_A represents herself as someone whose private assessment of her own capacity is accurate enough to justify taking the ambitious action. The social cost of later reinterpretation is therefore not only a material or reputational loss. It is also an epistemic loss: observers revise their beliefs about whether the sender correctly understood her own type.

The section makes three clarifications. First, the exposure index Φ_θ is a true-type-indexed primitive object, derived from primitive self-model-accuracy levels. It is not an equilibrium object generated by a previous separating history. Second, the relevant channel is adverse repricing, not truth verification. A high-precision audit would tend to confirm genuinely accurate high types; that is not the technology studied here. Third, the D1 argument later uses Φ_θ as a cost attached to the sender's true type, not as a cost attached to the receiver's off-path attribution. This keeps the exposure technology separate from the belief refinement that interprets deviations.

3.1 Belief-dependent exposure

Let $j \in \{L, H\}$ denote the current judgment-intensity state, with associated scrutiny multiplier $\mu_j > 0$. The low state L represents ordinary evaluation and the high state H represents intensified evaluation. The state variable j is not a political primitive. It is an exogenous state of the social evaluation technology.

After s_A , observers form beliefs not only about the sender's ability θ , but also about the accuracy of the sender's self-model. Write

$$m \in \{0, 1\}$$

for the event that the sender's self-model is accurate, where $m = 1$ denotes accurate self-assessment. The sender cares not only about what observers infer about θ , but also about what observers infer about the sender's own assessment of θ .

For a finite hierarchy of beliefs $(\mu^2, \dots, \mu^k)$, let

$$L(\mu^2, \dots, \mu^k; j, x)$$

denote the epistemic penalty in judgment state j and exposure state $x \in \{0, 1\}$. The exposure state records whether the sender previously chose the ambitious signal. Choosing s_A sets $x = 1$ next period; choosing s_P clears exposure next period.

The total one-period psychological payoff can be written as

$$u_\theta(s, j, x; \mu^2, \dots, \mu^k) = v_\theta(s) - c_\theta(s) - \mu_j L(\mu^2, \dots, \mu^k; j, x),$$

where $v_\theta(s)$ is the ordinary reputational or material payoff from the signal, $c_\theta(s)$ is the intrinsic cost of the signal, and the final term is the judgment-weighted epistemic penalty.

The next assumption is purely regularity. It rules out discontinuities inside a fixed judgment state. The discontinuities studied below will come from changes in the transition probability into high scrutiny, not from discontinuous psychological utility within a regime.

Assumption 2 (Regime-conditional continuity). *For each fixed $j \in \{L, H\}$ and $x \in \{0, 1\}$, the function*

$$L(\mu^2, \dots, \mu^k; j, x)$$

is continuous in the relevant finite hierarchy of higher-order beliefs. The action set, type space, and exposure state space are finite.

Lemma 1 (Regime-conditional continuity). *Under Assumption 2, for each fixed judgment-intensity state j , the induced psychological payoff is continuous in the relevant higher-order belief hierarchy. Hence the regime-conditional psychological game satisfies the standard continuity requirement for belief-dependent payoffs. On any open region where the relevant incentive constraints are strict, the associated pure-strategy equilibrium selection is locally constant.*

Proof. Fix j and x . The ordinary payoff terms $v_\theta(s)$ and $c_\theta(s)$ are defined on a finite action and type space. The multiplier μ_j is constant within the state j . By Assumption 2, the epistemic penalty $L(\mu^2, \dots, \mu^k; j, x)$ is continuous in the relevant belief hierarchy. Multiplication by the fixed scalar μ_j and addition of finite ordinary payoff terms preserve continuity. Thus, the regime-conditional psychological payoff is continuous.

Since the type and action spaces are finite, strict incentive inequalities are preserved under sufficiently small perturbations of beliefs and payoff parameters within a fixed state. Therefore, wherever the relevant incentive constraints are strict, the same pure-strategy profile remains selected in a neighborhood of the primitive. Any discontinuous change in equilibrium behavior must therefore come from a change in the relevant state, transition probability, or binding incentive constraint, rather than from a discontinuity of belief-dependent utility within the fixed state. $\square$

3.2 Primitive self-accuracy levels

The exposure index is built from a level channel, not a precision channel. Let

$$p_\theta := \Pr(m = 1 \mid \theta)$$

denote the primitive probability that a type- θ sender has an accurate self-model. The key ordering is

$$1 - \delta > p_H > p_L > \bar{p} > \delta > 0.$$

Higher types are genuinely more self-accurate in the primitive environment. The lower benchmark $\bar{p}$ is the assessment induced by adverse repricing under high scrutiny.

This assumption should not be read as signal precision. Precision is a fidelity concept: a more precise audit reveals the true state more accurately. If high types are genuinely more accurate, a fidelity-improving audit tends to confirm them, not punish them. The model instead studies adverse social repricing. In a high-scrutiny state, an exposed self-presentation can be reinterpreted downward with a positive probability that is not driven by the sender's true merit. The economic loss is increasing in the amount of self-model reputation-at-risk carried by the true type.

Let $\alpha \in (0, 1]$ be the probability, conditional on high scrutiny and exposure, that the ambitious self-presentation is adversely repriced. This probability is taken to be independent of type in the baseline. It can be normalized into μ_H , but keeping it visible makes the interpretation clear: high scrutiny is not a deterministic truth audit. It is a state in which exposed self-presentations face a type-independent risk of adverse reinterpretation.

The epistemic exposure index is

$$\Phi_\theta = \alpha D_{\text{KL}}(\text{Bern}(p_\theta) \parallel \text{Bern}(\bar{p})).$$

The direction of the divergence reflects the economic object. The loss is not the ordinary probability of failure. It is the collapse from a carried self-accuracy level to a lower adverse benchmark. Judgment intensity prices this exposure:

$$C_E(\theta, j) = \mu_j \Phi_\theta.$$

Thus Φ_θ is neither history-induced reputation capital nor signal precision. It is the true-type-indexed loss from adverse repricing of self-model exposure. The receiver's later off-path attribution may affect the reputational return to a deviation, but it does not redefine Φ_θ .

3.3 Bayesian-surprise complementarity

Lemma 2 (Bayesian-surprise foundation for scrutiny-type complementarity). *Suppose*

$$1 - \delta > p_H > p_L > \bar{p} > \delta > 0.$$

Define

$$\Phi_\theta = \alpha D_{\text{KL}}(\text{Bern}(p_\theta) \parallel \text{Bern}(\bar{p}))$$

with $\alpha > 0$, and let

$$C_E(\theta, j) = \mu_j \Phi_\theta.$$

Then the epistemic cost has increasing differences in type and judgment intensity. For every $\mu' > \mu$,

$$C_E(\theta_H, \mu') - C_E(\theta_H, \mu) > C_E(\theta_L, \mu') - C_E(\theta_L, \mu).$$

Equivalently, the marginal cost of higher judgment intensity is larger for the high-self-accuracy type.

Proof. For Bernoulli beliefs,

$$D_{\text{KL}}(\text{Bern}(p) \parallel \text{Bern}(\bar{p})) = p \log \frac{p}{\bar{p}} + (1 - p) \log \frac{1 - p}{1 - \bar{p}}.$$

Differentiating with respect to p gives

$$\frac{\partial}{\partial p} D_{\text{KL}}(\text{Bern}(p) \parallel \text{Bern}(\bar{p})) = \log \frac{p(1 - \bar{p})}{\bar{p}(1 - p)}.$$

This derivative is positive whenever $p > \bar{p}$. Since $p_H > p_L > \bar{p}$, it follows that

$$\Phi_H > \Phi_L.$$

Therefore, for every $\mu' > \mu$,

$$[C_E(\theta_H, \mu') - C_E(\theta_H, \mu)] - [C_E(\theta_L, \mu') - C_E(\theta_L, \mu)] = (\mu' - \mu)(\Phi_H - \Phi_L) > 0.$$

Thus higher judgment intensity raises the epistemic cost of the high-self-accuracy type more than that of the low-self-accuracy type. $\square$

Lemma 2 is the first reversal relative to standard signaling theory. In a Spence environment, high types are more willing to bear the ambitious signal because they face a lower ordinary cost of sending it. Here, high types also carry more self-model reputation-at-risk. When high scrutiny adversely reprices exposed self-presentation, the same type characteristic that supports ambitious action under low scrutiny becomes the source of larger exposure.

Remark 1 (Repricing, not precision). *The sign of Lemma 2 depends on the level channel. Replacing p_θ with signal precision would change the model. A higher-precision audit is a fidelity technology and tends to confirm an accurate high type. It therefore does not generate top-down withdrawal. The primitive needed here is a type-indexed self-accuracy level combined with an adverse repricing state, not a variance or precision parameter.*

3.4 From self-accuracy to crash risk

The preceding lemma converts primitive self-accuracy into exposure. A high type has more to gain from separation, but also more to lose from a future state that adversely reinterprets her ambitious signal as evidence about self-model risk.

Let B_θ denote the current net separation rent from choosing s_A rather than s_P in the low-judgment state, before future high-scrutiny exposure is priced. Let

$$q = \Pr(j_{t+1} = H \mid j_t = L)$$

be the transition probability into high scrutiny. If choosing s_A creates the one-period shadow, the low-state payoff gain from choosing s_A can be written as

$$D_\theta^L(q) = B_\theta - \beta q \mu_H \Phi_\theta.$$

The associated threshold is

$$q_\theta^* = \frac{B_\theta}{\beta \mu_H \Phi_\theta},$$

whenever $B_\theta > 0$.

Thus top-down withdrawal is governed not by exposure alone, but by the ratio between exposure and current rent:

$$\frac{\Phi_\theta}{B_\theta}.$$

The high type exits first precisely when

$$\frac{\Phi_H}{B_H} > \frac{\Phi_L}{B_L}.$$

In that case,

$$q_H^* < q_L^*.$$

As the probability of future high scrutiny increases, the high type's incentive constraint binds first.

Corollary 1 (Top-down threshold ordering). *Suppose $B_H > 0$, $B_L > 0$, and*

$$\frac{\Phi_H}{B_H} > \frac{\Phi_L}{B_L}.$$

Then

$$q_H^* < q_L^*.$$

Hence an increase in the transition probability into high scrutiny first destroys the ambitious-signal incentive of the type with the larger exposure-to-rent ratio.

Proof. The threshold is

$$q_\theta^* = \frac{B_\theta}{\beta \mu_H \Phi_\theta}.$$

Since $\beta\mu_H > 0$,

$$q_H^* < q_L^*$$

iff and only if

$$\frac{B_H}{\Phi_H} < \frac{B_L}{\Phi_L},$$

which is equivalent to

$$\frac{\Phi_H}{B_H} > \frac{\Phi_L}{B_L}.$$

□

The corollary gives the transition from the psychological exposure mechanism to the dynamic pooling result. The high type is not less capable of sending the ambitious signal. She is more exposed because her true type carries more self-model reputation-at-risk under adverse repricing.

4 Anticipatory Exposure and the D1 Flip

The previous section established two ingredients. First, conditional on a fixed judgment-intensity state, the belief-dependent payoff is continuous in the relevant higher-order belief hierarchy. Hence the psychological-game object is well-defined within each regime slice. Second, the epistemic exposure term is pinned to the sender's true type through primitive self-accuracy levels and adverse repricing. It is not a signal-precision parameter and not an equilibrium posterior.

This section shows how these ingredients change equilibrium selection. In standard costly signaling, the ambitious signal is credible because high types are most willing to send it. Here, sufficiently likely future scrutiny reverses that willingness ranking. The high type becomes less willing than the low type to carry the ambitious signal, because the high type has more self-model reputation-at-risk under adverse repricing. This reversal has two consequences. First, the separating profile fails when the high type's anticipatory incentive constraint binds. Second, in the pooling candidate, a D1-style forward-induction refinement no longer interprets an off-path ambitious signal as evidence of the high type.

4.1 The anticipatory incentive constraint

Let the current judgment-intensity state be low, denoted L . Let

$$q = \Pr(j_{t+1} = H \mid j_t = L)$$

be the transition probability into the high-scrutiny state. The agent chooses between the ambitious signal s_A and the pooling-safe signal s_P . Sending s_A creates an exposure state that can be repriced if the next-period state is H .

For type $\theta \in \{\theta_L, \theta_H\}$, let B_θ denote the current net separation surplus from sending s_A rather than s_P , before future high-scrutiny exposure is priced. This term includes the current reputational benefit of being interpreted as high type and subtracts current intrinsic and low-scrutiny epistemic costs. Let $\Phi_\theta > 0$ denote the true-type-indexed adverse-repricing exposure cost derived in Section 3. The low-state payoff gain from sending s_A is

$$D_\theta(q) = B_\theta - \beta q \mu_H \Phi_\theta,$$

where $\beta \in (0, 1)$ is the discount factor and μ_H is the high-scrutiny intensity.

The high type's separating incentive constraint is

$$D_H(q) \geq 0.$$

Thus define

$$q_{IC} = \frac{B_H}{\beta\mu_H\Phi_H},$$

assuming $B_H > 0$. For $q > q_{IC}$, the high type strictly prefers the pooling-safe signal s_P to remaining on the ambitious signal s_A , even if s_A is still interpreted as high type.

Lemma 3 (Anticipatory failure of separation). *Suppose the current state is L . If*

$$q > q_{IC},$$

then the separating profile

$$\theta_H \mapsto s_A, \quad \theta_L \mapsto s_P$$

is not a psychological equilibrium.

Proof. Under the separating profile, s_A is interpreted as the high-type signal. Hence the high type receives the current separation surplus B_H from s_A , but also internalizes the future exposure cost $\beta q\mu_H\Phi_H$. Its payoff gain from choosing s_A rather than s_P is therefore

$$D_H(q) = B_H - \beta q\mu_H\Phi_H.$$

If $q > q_{IC}$, then $D_H(q) < 0$. Thus the high type strictly prefers s_P to s_A . The proposed separating profile violates the high type's incentive constraint and therefore cannot be an equilibrium. $\square$

The point is not that the low type successfully mimics the high type. The separating profile collapses before that question arises. The high type exits because the expected future cost of adverse repricing exceeds the current benefit of being recognized.

4.2 The D1 flip

It remains to check whether pooling on s_P can be sustained. The possible obstruction is off-path resurrection: if all types pool on s_P , an off-path deviation to s_A might again be interpreted as evidence of high type, restoring the value of the ambitious signal.

This is where the refinement matters. Consider the pooling candidate in which both types choose s_P . The receiver observes an off-path signal s_A . Let a denote a receiver response, and let $R(a)$ be the reputational return from that response. For simplicity, assume $R(a)$ is type-independent and ordered by receiver favorability.

Let the cost of deviating to the ambitious signal for type θ be

$$K_\theta(q) = \bar{k}_\theta + \beta q\mu_H\Phi_\theta,$$

where $\bar{k}_\theta$ is the current cost of sending s_A in the pooling environment and $\beta q\mu_H\Phi_\theta$ is the anticipatory high-scrutiny exposure cost.

The exposure term Φ_θ is the deviator's true-type cost. D1 determines how the receiver interprets the off-path signal and therefore affects $R(a)$. It does not relabel the deviator's exposure technology. This separation is essential: otherwise the off-path belief used by the refinement would also redefine the cost object being compared.

Given receiver response a , type θ 's payoff gain from deviating to s_A is

$$G_\theta(a; q) = R(a) - K_\theta(q).$$

Define the profitable-deviation response set

$$\mathcal{D}_\theta(q) = \{a : G_\theta(a; q) > 0\}.$$

In the standard Spence ranking, the high type has a lower signaling cost, so

$$\mathcal{D}_L(q) \subset \mathcal{D}_H(q).$$

D1 then assigns the off-path ambitious signal to the high type. The present model reverses this inclusion when future scrutiny is sufficiently likely. By Lemma 2, high types have larger future exposure:

$$\Phi_H > \Phi_L.$$

Hence the high type's cost increases faster in q . The ranking flips when

$$K_H(q) > K_L(q),$$

or equivalently,

$$\bar{k}_H + \beta q \mu_H \Phi_H > \bar{k}_L + \beta q \mu_H \Phi_L.$$

Define

$$q_{D1} = \frac{\bar{k}_L - \bar{k}_H}{\beta \mu_H (\Phi_H - \Phi_L)},$$

whenever $\bar{k}_L > \bar{k}_H$. Then for $q > q_{D1}$,

$$K_H(q) > K_L(q).$$

Lemma 4 (D1 flip). *Suppose $\Phi_H > \Phi_L$ and $q > q_{D1}$. Then*

$$\mathcal{D}_H(q) \subsetneq \mathcal{D}_L(q).$$

Therefore, under the D1 criterion, an off-path ambitious signal s_A cannot be assigned to the high type. It must be assigned to the low type.

Proof. For any receiver response a , type θ 's gain from deviating to s_A is

$$G_\theta(a; q) = R(a) - K_\theta(q).$$

If $q > q_{D1}$, then $K_H(q) > K_L(q)$. Hence for every receiver response a ,

$$G_H(a; q) < G_L(a; q).$$

Therefore, whenever the deviation is profitable for the high type, it is also profitable for the low type:

$$G_H(a; q) > 0 \quad \Rightarrow \quad G_L(a; q) > 0.$$

Thus

$$\mathcal{D}_H(q) \subseteq \mathcal{D}_L(q).$$

The inclusion is strict whenever there exists a receiver response a such that

$$K_L(q) < R(a) < K_H(q).$$

Under D1, if one type can profitably deviate under a strictly wider set of receiver responses, the deviation is attributed to that type rather than to the type whose profitable-deviation set is smaller. Hence the off-path signal s_A is attributed to θ_L , not to θ_H . $\square$

The ambitious signal therefore changes its meaning. In the standard model, an off-path ambitious signal says: this must be a strong type, because only strong types can afford it. Here, when future scrutiny is sufficiently likely, the same signal says: this is unlikely to be the high type, because high types have the most to lose from carrying exposed ambition into future scrutiny.

4.3 Top-down pooling

We can now combine the two thresholds. The first threshold kills the separating profile by violating the high type's incentive constraint. The second threshold kills the off-path resurrection of the ambitious signal under D1.

Define

$$q^* = \max\{q_{IC}, q_{D1}\}.$$

Theorem 1 (Top-down pooling). *Suppose:*

1. $B_H > 0$, so the high type would choose the ambitious signal absent anticipatory high-scrutiny exposure;
2. $\Phi_H > \Phi_L$, as derived from primitive self-accuracy levels and Bayesian-surprise repricing in Section 3;
3. $q > q^* = \max\{q_{IC}, q_{D1}\}$;
4. the receiver's off-path beliefs satisfy the D1 criterion;
5. the analysis is restricted to the binary-signal pure-strategy class $\{s_A, s_P\}$.

Then the unique D1-consistent pure-strategy psychological equilibrium in this class is pooling on the safe signal:

$$\theta_H \mapsto s_P, \quad \theta_L \mapsto s_P.$$

Proof. First, since $q > q_{IC}$, Lemma 3 implies that the separating profile cannot be an equilibrium. Even if s_A is interpreted as the high-type signal, the high type strictly prefers s_P , because its expected future exposure cost exceeds its current separation surplus.

Second, consider the pooling candidate in which both types choose s_P . Since $q > q_{D1}$, Lemma 4 implies that the off-path ambitious signal s_A is attributed by D1 to the low type rather than to the high type. Therefore s_A carries no high-type reputational premium under D1-consistent beliefs.

Given this off-path belief, the high type has no profitable deviation to s_A . The deviation would create future exposure while failing to restore the high-type interpretation. The low type also has no profitable deviation: under the D1-consistent posterior, its deviation is interpreted as low type and therefore does not generate the reputational return needed to cover the cost of s_A .

Thus both types optimally choose s_P . Since separation has already been ruled out by the high type's incentive constraint, and the only off-path resurrection of s_A is ruled out by D1, the pooling profile is the unique D1-consistent pure-strategy psychological equilibrium within the binary-signal class. $\square$

The theorem identifies a reversal of the usual signaling logic. In standard costly signaling, separation fails from below when low types become able to mimic high types. Here, separation fails from above. The high type exits first because it has both the current separation rent and the largest future adverse-repricing liability.

Equivalently, the ambitious signal loses its credibility not because it becomes cheap for low types, but because it becomes too exposed for high types. Once future scrutiny is sufficiently likely, an agent who still sends the ambitious signal is no longer interpreted as confident. Under D1, it is interpreted as failing to internalize the future cost of exposure.

4.4 Interpretation

The result can be summarized by contrasting two mechanisms. In the standard Spence model, bad types cannot climb up. The high-cost signal is credible because only high types are willing to bear it. In the present model, good types climb down. The ambitious signal becomes dangerous precisely for the types that previously used it to separate.

This produces top-down pooling. The pooling outcome is not generated by low-type imitation. It is generated by high-type withdrawal. The signal s_A does not merely lose precision; its off-path meaning reverses. After the D1 flip, s_A is no longer evidence of high ability. It is evidence of low anticipatory discipline.

This is the belief-driven pooling reversal. Future scrutiny changes the sender's current incentives, and D1 changes the receiver's interpretation of deviations. Together they close the fixed point: high types withdraw because future scrutiny makes ambitious exposure costly, and once high types are expected to withdraw, ambitious deviations cease to be high-type credible.

5 Belief Loops and Fixed-Point Selection

Section 7 gives the observable shadow of withdrawal. Once scrutiny risk is high enough, the ambitious signal need not disappear; it can survive as a polarized object, carried by extreme high types and by agents with weak anticipatory discipline after ordinary high types have exited. That result is cross-sectional. It says what the ambitious signal looks like after withdrawal has started, but it does not yet show why withdrawal is a self-confirming equilibrium object.

The missing step is logically small but not vacuous. If high types are expected to remain on s_A , the signal carries a high-type premium. If high types are expected to withdraw, s_A becomes off path, and D1 can remove that premium. Thus the sender's participation decision and the receiver's interpretation of the signal must be solved together.

The local forces are already in place. Adverse repricing is pinned to true type, $\Phi_H > \Phi_L$ (Section 3). When the transition probability into high scrutiny is sufficiently large, D1 no longer assigns an off-path ambitious signal to the high type (Section 4). This section turns those local comparisons into a fixed-point selection result.

The point is not merely that separation fails and pooling survives. That would be a linear argument. The equilibrium object is circular. Whether s_A carries a high-type premium depends on whether high types are expected to occupy s_A . Whether high types occupy s_A depends on the premium they expect s_A to carry. The theorem below closes this loop.

5.1 Belief hierarchy and psychological assessments

The fixed point is taken over a psychological assessment, not over a strategy profile alone. At a low-scrutiny history h_t , let

$$\sigma_\theta(h_t) \in \{s_A, s_P\}$$

denote the type- θ sender's signal choice. Let

$$\rho_A(h_t) := \Pr(\theta_H \mid s_A, h_t)$$

denote the receiver's first-order posterior after observing the ambitious signal. If s_A is on path, ρ_A is pinned down by Bayes' rule. If s_A is off path, ρ_A is selected by the refinement.

The sender's payoff depends on what she expects the receiver to believe after s_A . Write

$$\hat{\rho}_{\theta,A}(h_t)$$

for the type- θ sender's second-order belief about the receiver's posterior following s_A . A psychological assessment is a triple

$$\mathcal{A} = (\sigma, \rho_A, \hat{\rho}_A),$$

where $\hat{\rho}_A = (\hat{\rho}_{H,A}, \hat{\rho}_{L,A})$. Belief consistency requires

$$\hat{\rho}_{\theta,A} = \rho_A \quad \text{for each } \theta \in \{\theta_H, \theta_L\}.$$

Thus the sender's second-order belief about the receiver's first-order posterior must agree with the posterior induced by the assessment itself.

Crucially, the fixed point is evaluated exactly at this second-order hierarchy layer. To formally embed this within the framework of psychological games [?], the sender's expected utility must be continuous with respect to the relevant belief hierarchy. Here, the net benefit $B_\theta(\hat{\rho}_{\theta,A})$ depends continuously on the expected first-order posterior. While the D1 refinement injects a discontinuous jump into the *receiver's* belief map $\rho_A(\sigma)$ when the signal becomes off-path, the fixed point is resolved on the composition of these expectations, ensuring that the belief loop strictly and continuously closes at $\hat{\rho}_{\theta,A} = \rho_A$.

This is the level at which the fixed point is closed. The exposure term Φ_θ remains true-type indexed. It is not changed by the receiver's posterior. The receiver's posterior changes the current reputational benefit of the signal; it does not relabel the deviator's adverse-repricing exposure.

Let

$$B_\theta(\rho)$$

denote the type- θ current net benefit from choosing s_A rather than s_P when the ambitious signal is interpreted with posterior ρ . This term includes the current reputational return from the receiver's interpretation, net of current signal cost and low-scrutiny epistemic cost. It excludes future high-scrutiny exposure. The fixed B_H used in Section 4 is the full high-type-attribution value:

$$B_H = B_H(1).$$

Assume

$$B_H(\rho) \text{ is weakly increasing in } \rho, \quad B_H(1) > 0, \quad B_H(0) \leq 0.$$

The first inequality says that a higher high-type posterior makes the ambitious signal more valuable. The last inequality says that an ambitious signal interpreted as low-type evidence carries no high-type premium.

For the low type, assume

$$B_L(0) < 0.$$

Thus if s_A is interpreted as low-type evidence, the low type does not receive enough current benefit to justify choosing it.

5.2 Depreciating exposure and the withdrawal threshold

The dynamic withdrawal motive requires exposure to be persistent but not permanently locked. A sender must be able to reduce future exposure by withdrawing from the ambitious signal. To make this explicit, let

$$e_t \geq 0$$

denote the sender's exposed stock at the beginning of period t . Sending s_A adds one unit of new exposed ambition; choosing s_P adds no new unit. Existing exposure depreciates at rate $\eta \in (0, 1]$:

$$e_{t+1} = (1 - \eta)e_t + \mathbf{1}\{s_t = s_A\}.$$

The binary exposure specification in Section 2 is the special case $\eta = 1$, where withdrawal immediately removes the reduced-form exposure state. The general stock formulation makes clear why withdrawal has dynamic value: it stops new exposure from being created and lets old exposure decay.

Let

$$\Gamma(q, \eta) > 0$$

denote the discounted marginal continuation price of adding one unit of exposed ambition in the current low-scrutiny state. In the one-period reduced form used in Section 4,

$$\Gamma(q, \eta) = \beta q \mu_H.$$

In the stock version, $\Gamma(q, \eta)$ is the expected discounted value of the future judgment intensities applied to the marginal unit of exposure:

$$\Gamma(q, \eta) = \mathbb{E}_q \left[\sum_{\tau \geq 1} \beta^\tau (1 - \eta)^{\tau-1} \mu_{j_t + \tau} \middle| j_t = L \right].$$

The maintained monotonicity condition is

$$\Gamma(q, \eta) \text{ is continuous and strictly increasing in } q.$$

This condition says that a higher probability of entering high scrutiny raises the continuation price of current exposure.

Given posterior ρ , the high type's payoff gain from choosing s_A rather than s_P is

$$\Delta_H(\rho, q) = B_H(\rho) - \Gamma(q, \eta) \Phi_H.$$

Define the high-type withdrawal threshold by

$$q_{IC} = \inf\{q : B_H(1) - \Gamma(q, \eta) \Phi_H < 0\}.$$

In the one-period reduced form, this collapses to the threshold in Section 4:

$$q_{IC} = \frac{B_H(1)}{\beta \mu_H \Phi_H}.$$

Lemma 5 (Monotone stopping). *Suppose $B_H(\rho)$ is weakly increasing in ρ , $\Phi_H > 0$, and $\Gamma(q, \eta)$ is continuous and strictly increasing in q . Then the high type's stay advantage $\Delta_H(\rho, q)$ is strictly decreasing in q . In particular, if $q > q_{IC}$, then*

$$\Delta_H(\rho, q) < 0 \quad \text{for every } \rho \in [0, 1].$$

Thus the high type's withdrawal region is an upper interval in q .

Proof. For fixed ρ ,

$$\Delta_H(\rho, q) = B_H(\rho) - \Gamma(q, \eta) \Phi_H.$$

Since $\Phi_H > 0$ and $\Gamma(q, \eta)$ is strictly increasing in q , $\Delta_H(\rho, q)$ is strictly decreasing in q . Since $B_H(\rho) \leq B_H(1)$ for every $\rho \in [0, 1]$, the inequality

$$B_H(1) - \Gamma(q, \eta) \Phi_H < 0$$

implies

$$B_H(\rho) - \Gamma(q, \eta) \Phi_H < 0$$

for every $\rho \in [0, 1]$. Hence, once $q > q_{IC}$, no receiver posterior can make the high type willing to stay on s_A . $\square$

Lemma 5 is the threshold structure needed for the dynamic interpretation. The stopping region is not fragmented because the only continuation object that changes with q is the marginal exposure price, and that price moves monotonically.

5.3 D1 and the off-path belief map

The next object is the receiver's belief after an off-path ambitious signal. In a pooling candidate on s_P , the signal s_A is not observed on path. The receiver must decide whether an off-path s_A is more plausibly sent by θ_H or by θ_L .

Let the off-path cost of sending s_A for type θ be

$$K_\theta(q) = \bar{k}_\theta + \Gamma(q, \eta)\Phi_\theta,$$

where $\bar{k}_\theta$ is the current off-path cost component and $\Gamma(q, \eta)\Phi_\theta$ is the continuation exposure cost. Given receiver response a , let $R(a)$ be the reputational return from that response. The deviation gain is

$$G_\theta(a; q) = R(a) - K_\theta(q),$$

and the profitable-deviation response set is

$$\mathcal{D}_\theta(q) = \{a : G_\theta(a; q) > 0\}.$$

Because $\Phi_H > \Phi_L$, the high type's off-path cost rises faster with q . Define

$$q_{D1} = \inf\{q : K_H(q) > K_L(q)\}.$$

For $q > q_{D1}$,

$$K_H(q) > K_L(q),$$

and therefore

$$\mathcal{D}_H(q) \subsetneq \mathcal{D}_L(q)$$

under the strict-response richness condition used in Lemma 4. D1 then assigns the off-path ambitious signal to the low type. Hence the D1-selected posterior is

$$\rho_A = 0.$$

The on-path and off-path belief map can therefore be summarized as follows. Let

$$z \in \{0, 1\}$$

denote whether the high type is expected to occupy the ambitious signal:

$$z = 1 \quad \text{means } \theta_H \text{ chooses } s_A,$$

while

$$z = 0 \quad \text{means } \theta_H \text{ withdraws to } s_P.$$

For $q > q_{D1}$, the receiver's belief about s_A is

$$\rho_A(z) = \begin{cases} 1, & \text{if } z = 1 \text{ and } s_A \text{ is the separating high-type signal,} \\ 0, & \text{if } z = 0 \text{ and } s_A \text{ is off path, selected by D1.} \end{cases}$$

The first line is Bayes' rule on the separating path. The second line is D1 off path.

5.4 The belief-loop operator

Given the belief map $\rho_A(z)$, the high type's participation response is

$$T_q(z) = \mathbf{1} \{B_H(\rho_A(z)) - \Gamma(q, \eta)\Phi_H \geq 0\}.$$

A high-type belief-loop fixed point is a value $z \in \{0, 1\}$ such that

$$z = T_q(z).$$

This is the self-referential equation. High-type participation determines whether s_A is on path. The path status of s_A determines the receiver's posterior. The receiver's posterior determines the current rent of s_A . That rent determines high-type participation.

Define

$$q^* = \max\{q_{IC}, q_{D1}\}.$$

Theorem 2 (Belief-loop fixed-point selection). *Suppose:*

1. $B_H(\rho)$ is weakly increasing in ρ , with $B_H(1) > 0$ and $B_H(0) \leq 0$;
2. $B_L(0) < 0$;
3. $\Phi_H > \Phi_L$, as derived from primitive self-accuracy levels and Bayesian-surprise repricing;
4. $\Gamma(q, \eta)$ is continuous and strictly increasing in q , with $\eta \in (0, 1]$;
5. $q > q^* = \max\{q_{IC}, q_{D1}\}$;
6. off-path beliefs satisfy D1;
7. the analysis is restricted to the binary-signal pure-strategy class.

Then the unique D1-consistent psychological assessment in this class has

$$\theta_H \mapsto s_P, \quad \theta_L \mapsto s_P.$$

Equivalently, the unique high-type belief-loop fixed point is

$$z^* = 0.$$

Proof. First consider the candidate in which the high type occupies the ambitious signal, $z = 1$. In the separating interpretation, s_A is on path for the high type, so Bayes' rule gives

$$\rho_A(1) = 1.$$

The high type's payoff gain from choosing s_A rather than s_P is

$$\Delta_H(1, q) = B_H(1) - \Gamma(q, \eta)\Phi_H.$$

Since $q > q_{IC}$, this expression is strictly negative. Hence

$$T_q(1) = 0.$$

Therefore $z = 1$ is not a fixed point.

Now consider the candidate in which the high type withdraws, $z = 0$. Then s_A is off path for the high type. Since $q > q_{D1}$, the D1 flip assigns an off-path s_A to the low type. Hence

$$\rho_A(0) = 0.$$

The high type's payoff gain from deviating to s_A is

$$\Delta_H(0, q) = B_H(0) - \Gamma(q, \eta)\Phi_H.$$

Since $B_H(0) \leq 0$ and $\Gamma(q, \eta)\Phi_H > 0$,

$$\Delta_H(0, q) < 0.$$

Thus

$$T_q(0) = 0.$$

Therefore $z = 0$ is a fixed point.

It remains to rule out pure-strategy profiles in which the low type occupies s_A . If only the low type chooses s_A , then s_A is on path as low-type evidence, so $\rho_A = 0$. The low type's gain is

$$B_L(0) - \Gamma(q, \eta)\Phi_L < 0,$$

because $B_L(0) < 0$ and $\Gamma(q, \eta)\Phi_L > 0$. Hence the low type does not wish to occupy s_A under low-type attribution.

If both types choose s_A , then s_A is pooling rather than separating. Let $\pi_0 \in (0, 1)$ be the prior probability of θ_H . Bayes' rule gives $\rho_A = \pi_0$. Since $B_H(\rho)$ is weakly increasing and $\pi_0 < 1$,

$$B_H(\pi_0) \leq B_H(1).$$

Because $q > q_{IC}$,

$$B_H(1) - \Gamma(q, \eta)\Phi_H < 0.$$

Therefore

$$B_H(\pi_0) - \Gamma(q, \eta)\Phi_H < 0.$$

The high type would deviate from pooling on s_A to the safe signal. Hence pooling on s_A is not a pure-strategy equilibrium.

The only remaining pure-strategy profile is pooling on s_P . Under that profile, s_A is off path. D1 gives $\rho_A = 0$, and the two deviation inequalities above show that neither type gains from deviating to s_A . Thus pooling on s_P is the unique D1-consistent psychological assessment in the binary pure-strategy class. $\square$

5.5 The closed loop

The theorem closes the loop in both directions. If high types are expected to stay, then s_A is read as high type:

$$z = 1 \Rightarrow \rho_A = 1.$$

But when $q > q_{IC}$, even full high-type attribution is not enough to offset continuation exposure:

$$\rho_A = 1 \Rightarrow T_q(1) = 0.$$

Thus the separating belief system destroys itself:

$$z = 1 \Rightarrow \rho_A = 1 \Rightarrow T_q(1) = 0 \neq z.$$

If high types are expected to withdraw, then s_A is off path. When $q > q_{D1}$, D1 assigns the off-path ambitious deviation to the low type:

$$z = 0 \Rightarrow s_A \text{ off path} \Rightarrow \rho_A = 0.$$

Once $\rho_A = 0$, the ambitious signal carries no high-type premium, while it still creates true-type exposure for a high-type deviator:

$$\rho_A = 0 \Rightarrow T_q(0) = 0.$$

Thus the withdrawal belief reproduces itself:

$$z = 0 \Rightarrow \rho_A = 0 \Rightarrow T_q(0) = 0 = z.$$

This is the belief loop. High types withdraw because exposed ambition is too costly. Their withdrawal makes s_A off path. D1 then removes the high-type attribution from s_A . Once that attribution is removed, the current rent from returning to s_A disappears. Withdrawal is therefore self-confirming.

5.6 No interior mixing in this section

The fixed-point theorem is a pure-strategy selection result. It does not use an interior mixing probability for the high type. This restriction is deliberate. In a binary model with no low-type background arrival at s_A , any strictly positive high-type mass on s_A would keep the posterior after s_A degenerate in the high type. Continuous signal dilution therefore cannot be generated by high-type mixing alone.

Interior mixing requires an additional object, such as low-type mimetic entry, noisy audience inference, trembles, or a continuous type distribution. We explicitly choose to scope out mixing in this section to demonstrate the exact conditions under which the purely top-down collapse occurs in the binary pure-strategy class. The interior mixed regime reemerges naturally as the *Polarization* phenomenon handled in Section 7 (and the continuous-type phase transition), where the signal informativeness continuously degenerates without dying entirely. The present section proves only the sharper fixed-point claim: above q^* , the only D1-consistent pure belief loop is full withdrawal from the ambitious signal and pooling on s_P .

6 Dynamic Forward Invariance

Section 5 establishes that in the binary class, the one-period object selects pooling for $q > q^*$. This section adds the missing microfounded dynamic state and asks whether endogenous accumulation can reverse that outcome.

6.1 Trust-capital law of motion

Let (C_t, E_t) be a two-dimensional endogenous state observed before j_t is realized.

- C_t is semantic trust capital created by repeated pooling-safe behavior.
- E_t is engagement reinforcement (habituation) accumulated from sustained non-ambitious actions.

The state evolution is

$$\begin{aligned} C_{t+1} &= (1 - \rho_C)C_t + \eta_C \mathbf{1}\{s_t = s_P\} - \psi_C \mathbf{1}\{s_t = s_A\} - \xi_t^C, \\ E_{t+1} &= (1 - \rho_E)E_t + \eta_E \mathbf{1}\{s_t = s_P\} + \eta_A \mathbf{1}\{m_t = 1\} - \xi_t^E, \end{aligned}$$

where $m_t \in \{0, 1\}$ is a cheap trust-restoring message available to both types and ξ_t^C, ξ_t^E are bounded asynchronous shocks. This keeps the mechanism endogenous while preserving one-period exposure at the signaling margin.

Define a temporary semantic penalty term

$$\Sigma_t = \omega_C C_t + \omega_E E_t \geq 0.$$

Then, in low-scrutiny state L , the low-state advantage of sending ambitious is

$$\Delta_\theta^{\text{dyn}}(q_t, \lambda_t) = B_\theta(\lambda_t) - \beta q_t \mu_H \Phi_\theta - \Sigma_t,$$

where the current posterior λ_t appears in B_θ as in Section 5. Sending s_P avoids the future repricing term and does not create today's semantic penalty.

6.2 Unpriced trust-restoring messages

Cheap anthropomorphic messages are unpriced: the sender can choose $m_t = 1$ with no direct signal-dependent cost and no type restriction. Because both types can choose m_t identically, the induced change in (C_t, E_t) is type-neutral. Therefore these messages are shadowable.

To see dynamic single crossing, let receiver response a induce continuation return $R(a)$ from choosing s_A . The high- and low-type off-path gains are

$$G_H(a; t) = R(a) - \bar{k}_H - \beta q_t \mu_H \Phi_H - \Sigma_t, \quad G_L(a; t) = R(a) - \bar{k}_L - \beta q_t \mu_H \Phi_L - \Sigma_t.$$

The term Σ_t cancels in the comparison, so adding or relaxing trust-capital does not alter the type comparison. If $q_t > q_{D1}$, then $G_H(a; t) < G_L(a; t)$ for every receiver response, so s_A is still D1-attributed to the low type.

6.3 Dynamic Spence ratio and wedges

For any endogenous state (C_t, E_t) define the state-adjusted top-down ratio

$$\tilde{q}_\theta^{\text{IC}}(t) = \frac{B_\theta(\lambda) + \Sigma_t}{\beta \mu_H \Phi_\theta}.$$

Endogenous trust-capital terms appear symmetrically in numerators and therefore do not change the fundamental type-ordering created by (Φ_H, Φ_L) and static primitives. They shift the threshold pair, but not the sign of the top-down reversal.

A true restoration requires an externally priced type-specific wedge, for example a direct penalty or premium that depends on sender type/claim context and is not a pure continuation by-product of C_t and E_t .

6.4 Forward invariance

Let

$$\mathcal{Q} := \{q \in [0, 1] : q \geq q^*\}, \quad q^* = \max\{q_{\text{IC}}, q_{\text{D1}}\}.$$

Assume that under pooling on s_P , the process satisfies:

1. $\Sigma_{t+1} - \Sigma_t \geq \underline{d} - \xi_t$ for some $\underline{d} > 0$ and bounded $\xi_t \in [0, \bar{\xi}]$;
2. $\Sigma_0 \geq 0$ and $\bar{\xi} < \underline{d}$;

3. $\beta q \mu_H \Phi_H - B_H(0) - \Sigma_0 > 0$ for all $q \in \mathcal{Q}$.

Theorem 3 (Pooling interval forward invariance). *Under these assumptions and strict D1 preferences at $q \in \mathcal{Q}$, if period t is in the pooling assessment of Section 5, then period $t + 1$ is also in that assessment with probability one. Equivalently, $\mathcal{Q}$ is almost surely forward-invariant.*

Proof. In the pooling assessment, $\lambda_t = 0$ for any off-path ambitious deviation. The high-type deviation margin is

$$\Delta_H^{\text{dyn}}(q_t, 0) = B_H(0) - \beta q_t \mu_H \Phi_H - \Sigma_t.$$

By assumption 3 and $\Sigma_t \geq \Sigma_0$, $\Delta_H^{\text{dyn}}(q_t, 0)$ is negative for all $q_t \in \mathcal{Q}$. Assumption 1 says endogenous drift in Σ_t dominates bounded asynchronous noise, so this negativity cannot be overturned by one-period shocks. Hence high type never has a profitable ambiguous-ambitious deviation from s_P once in the pooling band, and D1 therefore continues to map s_A to low attribution. The same argument applies period-by-period by induction. $\square$

The conclusion is defensive. Dynamic trust-capital and habituation can be modeled explicitly and can affect local margins, but they do not create a new stopping phenomenon. They do not by themselves repair top-down collapse.

6.5 Population-density shape in the critical linearized system

For intuition (and to avoid stale wording from prior drafts), consider the linearized density

$$\mathcal{L}_\kappa \varphi = \lambda \varphi$$

for the trust-state law under pooling. The relevant object is its principal eigenpair (λ_1, φ_1) ; by definition $\varphi_1 \geq 0$ on the domain, not sign-changing. In the parameter regime of interest, the population profile is therefore single-peaked: it starts from the boundary with a left foot at $x = 0$, peaks in the safe region, and after the turning point decays into the dangerous region with exponential tail. A collapse-zone oscillation is not part of φ_1 ; that oscillatory sine pattern belongs to non-principal local approximations and is not a valid population density.

Airy appears only as a local normal form near the turning point, where the reduced equation takes the form

$$\phi''(x) - x \phi(x) = 0,$$

so $\phi(x) \propto \text{Ai}(x)$ for matching. Elsewhere, φ_1 governs the full profile.

At $\lambda = 0$, the critical transition, imposing the Dirichlet condition at the left boundary yields the principal-root condition: the boundary is pinned at the first zero of the matching profile. This is not fine tuning; it is the algebraic consequence of the critical principal-eigenvalue condition together with the boundary constraint. The earlier “coincidence” wording should therefore be replaced by this inference.

7 Observable Implications: Polarization of Ambitious Signals

The binary model delivers a sharp equilibrium statement: once the transition probability into high scrutiny exceeds the critical threshold, the high type exits the ambitious signal and pooling on the safe signal is selected under D1-consistent beliefs. This section records the observable implication in a richer type space. Near the threshold, the quality of the ambitious signal need not decline smoothly. The more distinctive prediction is polarization: ordinary winners withdraw, while low-calibration agents and extreme high types may remain.

7.1 Continuous types

Let the type space be an interval

$$\Theta = [\underline{\theta}, \bar{\theta}].$$

The sender chooses between s_A and s_P . The current low-scrutiny net separation rent from s_A is $B(\theta)$. The true-type-indexed adverse-repricing exposure is $\Phi(\theta)$, derived from primitive self-accuracy levels

$$p(\theta) = \Pr(m = 1 \mid \theta), \quad p'(\theta) > 0,$$

through

$$\Phi(\theta) = \alpha D_{\text{KL}}(\text{Bern}(p(\theta)) \parallel \text{Bern}(\bar{p})).$$

The low-state payoff gain from the ambitious signal is

$$D(\theta, q) = B(\theta) - \beta q \mu_H \Phi(\theta).$$

Thus type θ chooses s_A exactly when

$$q < q^*(\theta), \quad q^*(\theta) = \frac{B(\theta)}{\beta \mu_H \Phi(\theta)}.$$

The binary theorem corresponds to the case in which the top type's threshold is crossed first. In the continuous model, the entire shape of $q^*(\theta)$ determines which types remain on the ambitious signal.

7.2 Hollowing out

A simple mean-reversion story would predict that, as scrutiny risk rises, the ambitious signal gradually becomes lower quality. That is not the generic implication of top-down pooling. The model removes types according to the exposure-to-rent ratio

$$\frac{\Phi(\theta)}{B(\theta)},$$

not according to type itself.

Suppose $q^*(\theta)$ is nonmonotone: it is low on an intermediate-high region of the type space but rises again in the extreme upper tail. Economically, these are environments in which ordinary winners carry substantial reputation-at-risk but do not have enough surplus to dominate future adverse repricing, while extreme types have sufficiently large current surplus or sufficiently robust self-models to remain exposed.

For a given q , the set of types choosing the ambitious signal is

$$A(q) = \{\theta \in \Theta : q < q^*(\theta)\}.$$

If $q^*(\theta)$ has an interior trough, then for transition probabilities in the corresponding range,

$$A(q) = [\underline{\theta}, \theta_1(q)] \cup [\theta_2(q), \bar{\theta}]$$

for some $\theta_1(q) < \theta_2(q)$. The ambitious signal is then generated by two separated groups: low or poorly calibrated types whose exposure-to-rent ratio is low because they have little to lose, and extreme high types whose rents remain large enough to bear exposure.

Proposition 1 (Polarized degradation). *Suppose $q^*(\theta)$ is continuous and there exist $\theta_1 < \theta_2$ such that*

$$q^*(\theta) > q \quad \text{for } \theta \in [\underline{\theta}, \theta_1] \cup [\theta_2, \bar{\theta}]$$

and

$$q^*(\theta) < q \quad \text{for } \theta \in (\theta_1, \theta_2).$$

Then the type distribution conditional on the ambitious signal has support on two separated intervals. In particular, the ambitious signal is not a monotone high-type signal in this region.

Proof. The sender chooses s_A if and only if $q < q^*(\theta)$. The stated inequalities imply that this condition holds exactly on $[\theta, \theta_1] \cup [\theta_2, \bar{\theta}]$ and fails on (θ_1, θ_2) . Hence the support of θ conditional on observing s_A is the union of two separated intervals. This support is not an upper set in Θ . Therefore s_A is not a monotone high-type signal. $\square$

The observable shadow is a hollowing-out of credible ambition. The conditional mean of type among ambitious senders may fall,

$$\mathbb{E}[\theta \mid s_A] \downarrow,$$

but the more distinctive prediction is a rise in dispersion:

$$\text{Var}(\theta \mid s_A) \uparrow$$

relative to the pre-threshold separating region. The surviving ambitious signal can be generated by extreme competence or by poor anticipatory discipline. This is why mean reversion is too weak a description. The ambitious signal becomes polarized.

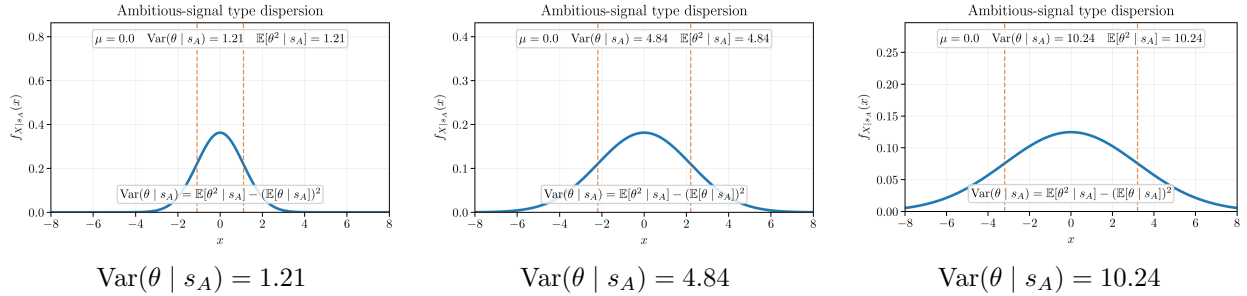


Figure 1: Model simulations: increasing conditional dispersion for the ambitious signal.

7.3 Relation to the D1 flip

The polarization result is the observable analogue of the D1 flip in Section 4. In the binary model, once $q > q_{D1}$, the off-path ambitious signal is attributed to the low type because the high type has the smaller profitable-deviation response set. In the continuous model, the same force makes the ambitious signal nonmonotone. It no longer means simply “high type.” It can mean either extreme surplus or low anticipatory discipline.

The epistemic exposure term remains pinned to true type throughout this argument. The receiver’s interpretation affects the reputational return to s_A , while $\Phi(\theta)$ describes the true-type cost of carrying exposed self-presentation into a possible adverse repricing state. This separation prevents the off-path belief system from redefining the underlying exposure technology.

7.4 Empirical language

The model predicts a specific kind of degradation. It is not merely that ambitious signals become noisier. It is that the middle of the credible ambitious class exits first. Agents with too little reputation-at-risk may continue because adverse repricing cannot take much from them. Agents with overwhelming surplus may continue because the ambitious signal remains worthwhile. Agents in between withdraw because they have enough to lose and not enough surplus to dominate the future exposure cost.

In the standard signaling model, the high signal is protected because bad types cannot climb up. In this model, the high signal is hollowed out because good types climb down. The remaining signal can look

extreme: true outliers and badly calibrated imitators share the same visible action. That polarization is the observable signature of top-down pooling.

The polarization result identifies the cross-sectional shadow of withdrawal. What remains is to connect that shadow to the dynamic incentive that makes withdrawal rational before high scrutiny is realized. Section 6 supplies this step by showing that the same high-type IC threshold can be interpreted through forward invariance. Section 5 provides the fixed-point closure in the binary core.

8 Continuous-Time Population Dynamics and the Top-Down Collapse

While Section 7 established the static conditions for the unraveling of the separating equilibrium, the observable hollowing-out of the signal is a macroscopic demographic phenomenon. In reality, the breakdown of the single-crossing property does not instantly evaporate the high-type population. Instead, it triggers a continuous-time demographic evacuation. To formally characterize the resulting cross-sectional density of types, we embed the discrete stage-game into a spatial population model.

The goal of this section is to prove that the exact threshold q_{IC} where the discrete single-crossing property fails reappears precisely as the macroscopic turning point in the continuous density equation, governing a phase transition from a persistent population to an exponential vacuum.

8.1 Continuous State Space and Demographic Exit

Let $x \in [0, \infty)$ denote a continuous state variable capturing “proximity to power” or observable status level. The scrutiny mapping $q(x)$ describes the probability of encountering the adverse repricing state at status x . The fundamental feature of this malicious scrutiny regime is that visibility attracts risk:

$$\frac{dq(x)}{dx} > 0.$$

The higher an agent’s status, the higher the probability of adverse repricing.

In the discrete game, the type- H sender evaluates the one-period stay advantage of maintaining the ambitious signal s_A :

$$V_H(q) = B_H(1) - \beta q \mu_H \Phi_H,$$

where $B_H(1)$ is the reputational premium of being read as a high type, and the second term is the expected penalty of epistemic exposure. The condition $V_H(q_{IC}) = 0$ marks the exact breakdown of the single-crossing property.

In the continuous-time population model, high-type agents diffuse across the status space with variance σ^2 . Crucially, an agent’s decision to withdraw from s_A to s_P is not a spatial movement along x , but a *demographic exit* from the ambitious pool. The local net exit rate is directly proportional to the stage-game disadvantage: $k(x) \propto -V_H(q(x))$.

Therefore, the cross-sectional density of high types $\rho(x, t)$ in the ambitious pool satisfies a reaction-diffusion equation (a spatial birth-death process):

$$\frac{\partial \rho}{\partial t} = \frac{\sigma^2}{2} \frac{\partial^2 \rho}{\partial x^2} + V_H(q(x)) \rho.$$

In regions where $V_H > 0$, the signal yields a net surplus, attracting entry or retention (acting as a “birth” rate). Where $V_H < 0$, the exposure penalty overwhelms the premium, driving demographic exit (a “killing” rate).

8.2 Stationary Density and the Turning Point Weld

We look for a long-run stable cross-sectional density $\rho(x)$ by setting $\partial\rho/\partial t = 0$. The stationary density equation becomes:

$$-\frac{\sigma^2}{2}\rho''(x) - V_H(q(x))\rho(x) = 0. \quad (1)$$

Let the effective resistance potential be $V(x) \propto -V_H(q(x))$. The equation reduces to a standard second-order boundary value problem: $-\sigma\rho''(x) + V(x)\rho(x) = 0$.

Because the scrutiny probability $q(x)$ is strictly increasing in x , the one-period stay advantage $V_H(q(x))$ is strictly decreasing. Therefore, the effective potential $V(x)$ is strictly increasing. It crosses zero from negative to positive at a critical spatial threshold x_c .

Theorem 4 (The Turning Point Weld). *The physical turning point x_c of the stationary density equation perfectly coincides with the discrete IC breakdown threshold:*

$$V(x_c) = 0 \iff q(x_c) = q_{IC}.$$

Furthermore, $V'(x_c) > 0$.

This theorem confirms that the turning point is not an assumed parameter in a differential equation, but the direct macroscopic manifestation of the discrete single-crossing failure. The boundary x_c separates the state space into two distinct regimes:

- **The Safe Zone** ($x < x_c$): Here $V(x) < 0$. The density $\rho(x)$ is locally concave ($\rho'' < 0$), allowing population mass to accumulate.
- **The Collapse Zone** ($x > x_c$): Here $V(x) > 0$. The density $\rho(x)$ is locally convex ($\rho'' > 0$), forcing the population to decay.

8.3 The Critical Snapshot and Phase Transition

To physically ground the density function, we require that the population density must be non-negative everywhere ($\rho(x) \geq 0$) and bounded at the protected limit (zero proximity to power). In continuous-time macroeconomics, this translates to anchoring the left boundary $\rho(0) = 0$ (Dirichlet condition) and discarding any explosive roots at infinity (the transversality condition).

Because $V(x)$ is locally linear around x_c , equation (??) is locally governed by the Airy equation. The unique non-explosive solution transitioning from the safe zone into the collapse zone takes the form of the first-kind Airy function, $\text{Ai}(\cdot)$.

The stationary state of the population is governed by the principal eigenvalue λ of the operator $-\sigma\partial_{xx} + V(x)$. As the baseline scrutiny parameter shifts, the system undergoes a macroscopic phase transition:

1. **Subcritical** ($\lambda > 0$): When baseline scrutiny is low ($q \ll q^*$), the turning point x_c is pushed far to the right. The safe zone is vast, and the high-type population is persistent and robust. The Spence signaling mechanism functions identically to the standard textbook model.
2. **Critical Snapshot** ($\lambda = 0$): As scrutiny rises, the turning point x_c intrudes into the core population zone. Figure ?? illustrates this exact critical moment. The density forms a single hump anchored in the safe zone. Crucially, as x crosses x_c , the curve passes through an inflection point and plunges into an *exponential vacuum*. The highest-status positions ($x > x_c$) are physically hollowed out. The zero-boundary condition at $x = 0$ mathematically coincides exactly with the first root of the Airy function, satisfying the criticality condition without internal oscillations.

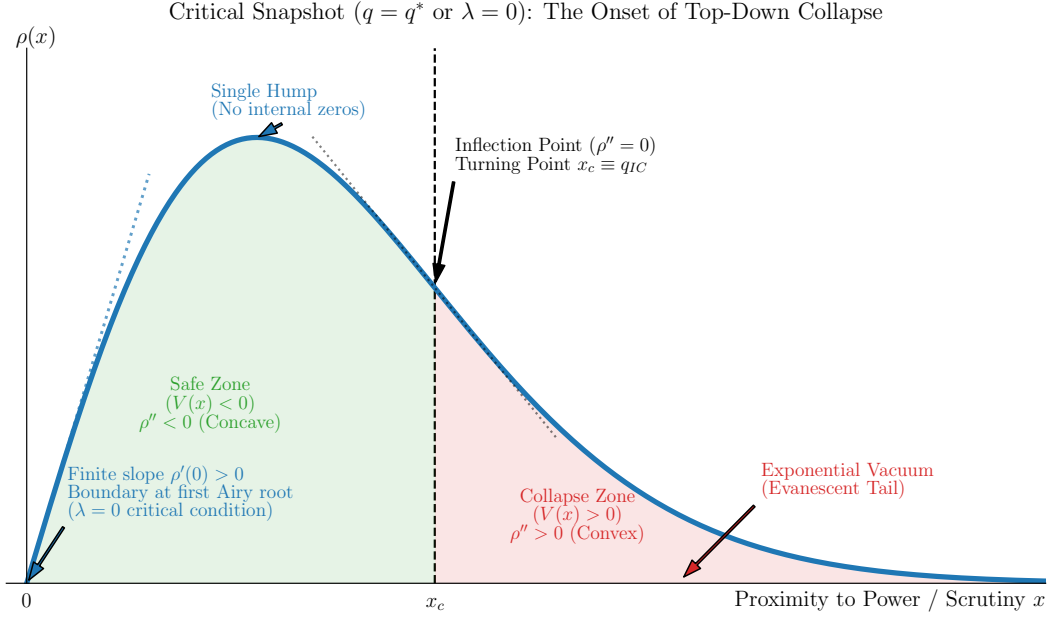


Figure 2: The critical snapshot at $q = q^*$ (or $\lambda = 0$). The stationary cross-sectional density $\rho(x)$ forms a single hump anchored at the protected boundary ($x = 0$), representing the accumulation of high types in the safe zone. As proximity to power surpasses the critical threshold x_c , the curve undergoes an inflection and rapidly hollows out into an exponential vacuum. This spatial phase transition is the continuous macroscopic manifestation of top-down pooling.

3. **Supercritical Collapse** ($\lambda < 0$): When scrutiny surpasses the threshold ($q > q^*$), the principal eigenvalue turns negative. The demographic exit rate strictly dominates any diffusion influx across the entire domain. *No stationary density exists*. The population is entirely evacuated, and top-down pooling is complete.

The macroscopic hollowing-out is therefore the continuous shadow of the discrete D1 flip. The top types do not merely vibrate in a noisy equilibrium; they disappear into an exponential vacuum created by their own epistemic exposure.

References